

What Are We Measuring? Benchmark Integrity and Zone-Based Reframing for Emotion Recognition in Autistic Children

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Abstract—Automated emotion recognition is widely proposed as an assistive technology for autistic children, and recent papers report up to 97.95% accuracy on public autism facial-expression benchmarks. Independently collected clinical corpora report 40–78% on the same task. We audit the public archive behind many of the former figures. Re-acquiring it, we find it ships two overlapping dataset trees sharing 45% of the smaller tree’s images, that 21.4% of its files are byte-identical duplicates, and that 119 duplicate groups carry contradictory emotion labels on pixel-identical images. Holding features, classifier, and images fixed and varying only the split protocol, five-category balanced accuracy spans 0.346–0.750 across five defensible readings of the same archive. Duplicate leakage is not the main driver: a nearest-neighbour classifier that memorises training data exactly gains only 2.1 points from a split where 80.6% of test images have a family member in training, and reaches 0.754 at best, so contamination cannot produce the published figures. The label contradictions are sharply asymmetric: 279 of 280 zone-level conflicts fall in negative affect, against 59.5% expected under a permutation null preserving class marginals ($p < 0.0005$). The archive’s labelling is far more consistent than chance overall yet nearly all of its residual inconsistency lies within negative valence. Collapsing to three action-oriented Emotion Zones cuts chance-corrected protocol variance by 29.3%. We argue for coarser labels and multimodal context over finer face-only categories. This is a measurement audit, not a deployable classifier, and not a claim about any child’s internal state.

Index Terms—benchmark integrity, data leakage, label noise, affective computing, autism

I. INTRODUCTION

Roughly 1 in 31 eight-year-olds in the United States is diagnosed with autism spectrum disorder (ASD) [1]. Because emotional expression in autistic people is frequently misread by non-autistic observers—the *double empathy problem* [2]—automated emotion recognition is repeatedly proposed as a communication aid.

The literature on this task is in an unusual state. Papers evaluating on public autism facial-expression archives report accuracies from 80.0% [3] to 97.95% [4]. Papers evaluating on independently collected, consented clinical corpora report far

less: 40% for a commercial analyser on the EMBOA autism corpus [5], 57.95–64.92% for four commercial classifiers on parent-reported autistic children [6], and 66.43% for a classifier trained on typically developing children and tested on autistic children [7]. The gap tracks how the evaluation data were obtained rather than what the models are.

That correlation has become hard to ignore. Publishers have retracted work built on scraped autism face datasets, including the source paper for the archive we audit [8] and a facial emotion recognition paper [9], on the grounds that images were collected from the internet without documented diagnosis, ethical oversight, or guardian consent. That established a *provenance* failure. It did not establish whether the reported numbers were also measurement artifacts.

This paper supplies that measurement. Our contributions are:

- 1) A quantified integrity audit of a widely used public autism emotion archive, computed from the vendor’s own directory layout (Sec. III).
- 2) A protocol ablation holding features, classifier, and images fixed while varying only the split, plus a nearest-neighbour memorisation ceiling that bounds what contamination can produce (Sec. IV).
- 3) A permutation test showing the label contradictions concentrate in negative affect far beyond what class marginals predict (Sec. V).
- 4) A resulting argument for coarser action-oriented zones and multimodal context over finer face-only categories (Sec. VI).

We audit an instrument rather than build one. We make no claim about any child’s internal state, and we treat every label as a filing decision about an image rather than as ground truth about a person.

II. RELATED WORK

Cross-population transfer is established. Grossard *et al.* [7] ran the controlled experiment on lab-collected video from 157 typically developing and 36 autistic children with subject-independent cross-validation, obtaining 82.05% within-population against 66.43% across. Gaya-Morey *et al.* [10] repeated the design with twelve CNNs on an adjacent population. We do not claim this finding; we ask whether the public benchmarks now used to measure it can support the measurement.

Benchmark contamination is a mature field. Kapoor and Narayanan [11] catalogue eight leakage types across 294 affected papers. Barz and Denzler [12] found 3.3% and 10% duplicate rates in CIFAR test sets. Northcutt *et al.* [13] showed test-set label errors reorder model rankings, and Recht *et al.* [14] showed that the choice of evaluation set alone moves reported accuracy substantially. Ramos *et al.* [15] give the taxonomy we adopt: *hard* leakage for identical images across a split, *soft* for near duplicates, *intra-dataset* for overlap within one dataset’s splits. Adimoolam *et al.* [16] name *augmented duplicates*; Apicella *et al.* [17] formalise *synthesis leakage*, where train and test items derive from a common source under a generating transform. None of this work examines a clinical affect dataset.

Contamination in this archive has been asserted, not measured. The authors of the FERAC derivative [18] reported “class overlap and image duplication” here, removed duplicates, and dropped two classes as unrecoverable. They reported no detection method, threshold, duplicate rate, leakage rate, or effect of cleaning. Two groups have now flagged the same archive; what remains open is the quantity.

Dataset critique and documentation. Birhane and Prabhu [19] audit large scraped image corpora containing people, including minors, and argue such datasets are ethically defective at the point of collection. Gebru *et al.* [20] and Paullada *et al.* [21] propose the documentation practices whose absence we observe here.

Why the labels might be the problem. Barrett *et al.* [22] challenge the premise that emotional state is readable from facial configuration at all. Keating *et al.* [23], from over 265 million motion-capture points, find autistic and non-autistic expressions differ in which muscles carry each emotion, concluding the groups may be “essentially speaking a different language.” If expressions are ambiguous under a non-autistic reading, forcing them into six mutually exclusive categories should produce contradictions concentrated where categories are closest.

III. THE ARCHIVE

We audit the public Kaggle archive associated with [8].¹ All 1,672 SHA-256 digests recorded in an earlier run of ours are present in the fresh download, so the audit and the models in Sec. IV describe the same bytes. Statistics below come from the vendor’s own directory layout, before any re-splitting: this is what a downstream user receives.

Two overlapping datasets ship as one. The archive contains two top-level trees (Table I) with different class balances and different splits, sharing 323 unique images—45.0% of the smaller tree. This plausibly explains a puzzling feature of the literature: published descriptions of this dataset report its size

¹Slug `fatmamtalaat/autistic-children-emotion-ma-m-talaat`. We name it for reproducibility while noting it distributes images of minors of disputed provenance; readers should weigh that before downloading.

TABLE I
TWO DATASET TREES SHIPPED IN ONE ARCHIVE.

Tree	Files	Unique	Train	Test
Balanced copy	1,425	1,380	1,199	226
Imbalanced copy	833	717	758	75
Shared unique images: 323 (45.0% of smaller tree)				

variously as 348, 830, 1,603, and “2500+” images, consistent with different papers loading different trees.

One file in five is a duplicate. Of 2,258 labelled images only 1,774 are distinct by SHA-256; 484 files (21.4%) are byte-identical copies, in groups reaching size 5. Perceptual hashing at Hamming distance 4 adds 163 near-duplicate components. A filename suffix (`_aug_N`) identifies 1,020 augmentation families, 335 larger than one, the largest containing 9, so a substantial part of the archive is synthetic variation on fewer source images—synthesis leakage in the sense of [17]. Notably these families are identifiable from filenames alone, so the duplication was detectable without any image analysis.

Pixel-identical images carry contradictory labels. Under the vendor’s six categories, 119 byte-identical groups (327 files) contain members filed under more than one emotion. Establishing this requires no external ground truth and no annotator panel, only a hash: it is an internal inconsistency of the artifact, which is a cheaper and stronger claim than label-error detection in the sense of [13].

IV. HOW MUCH OF THE NUMBER IS THE PROTOCOL?

Contamination matters only if it moves results. We hold the images, the features, and the classifier fixed and vary *only* how train and test are separated. Features are frozen ViT-B/16 embeddings [24]; the head is a one-hidden-layer MLP with epoch-wise early stopping; every protocol runs over ten seeds [25]. Because *surprise* has no zone, it is dropped from *both* arms, so the categorical arm is a five-way task over exactly the images the zone arm sees and the two columns differ only in label granularity.

We add a 1-nearest-neighbour classifier as a *memorisation ceiling*. A low-capacity probe cannot exploit a leaked duplicate, so on its own it cannot tell us what an end-to-end fine-tuned network might extract from a contaminated split. 1-NN is the opposite extreme: it stores the training set exactly and recovers any byte-identical test twin with certainty, bounding the contamination effect from above without training anything.

Table II and Fig. 1 support three claims and refute a fourth we expected to make.

The reported number is largely a choice. Across five defensible protocols on identical images, five-category balanced accuracy spans 0.346 to 0.750—a 40.4-point range attributable entirely to how the archive is read. Seed dispersion is 2.3–6.3 points, so protocol dominates seed variance by roughly an order of magnitude. A reader given one number cannot tell where in that band it sits.

TABLE II

SAME IMAGES, SAME FEATURES, SAME CLASSIFIER; ONLY THE SPLIT CHANGES. “GRP” IS THE SHARE OF TEST IMAGES WHOSE DUPLICATE OR AUGMENTATION FAMILY ALSO APPEARS IN TRAINING. TEN SEEDS; SD OVER SEEDS.

Protocol	Bal. acc.	sd	1-NN	Grp	<i>n</i>
<i>Five categories</i> (chance = 0.200)					
Vendor split, imbal. tree	0.346	.050	0.348	26.1%	69
Train one tree, test other	0.500	.043	0.476	24.6%	69
Vendor split, bal. tree	0.558	.037	0.585	33.3%	189
Group-aware	0.700	.052	0.734	0.0%	180
Pooled random	0.750	.028	0.755	80.6%	293
<i>Three zones</i> (chance = 0.333)					
Vendor split, imbal. tree	0.497	.063	0.392	26.1%	69
Vendor split, bal. tree	0.562	.025	0.552	33.3%	189
Train one tree, test other	0.665	.025	0.661	24.6%	69
Group-aware	0.727	.034	0.733	0.0%	180
Pooled random	0.735	.023	0.736	80.6%	293

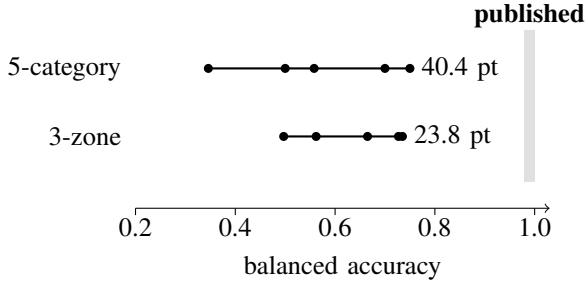


Fig. 1. Every dot is the same images, features, and classifier under a different split protocol. The spread is the protocol, not the model. No protocol—including a 1-NN memoriser on the most contaminated split—comes within 22 points of the published band.

Zones are markedly more protocol-stable. The same protocols span 23.8 points under the zone collapse against 40.4 under five categories. Because the two label spaces have different chance floors, we compare them chance-corrected as $(BA - 1/k)/(1 - 1/k)$: the ranges are 0.357 and 0.505, a **29.3%** reduction in protocol-induced variance. The coarser target is not merely easier; it is less sensitive to a methodological choice the reader cannot see.

Duplicate leakage is real but second-order. Pooled random leaks the family of 80.6% of its test images and exceeds the group-aware split by 5.0 points under five categories and 0.8 under zones. The memorisation ceiling is stricter still: 1-NN gains only 2.1 and 0.3 points respectively from the same contamination.

Contamination does not explain the published figures. The highest value any protocol reaches, under any model including the explicit memoriser, is 0.755. That is 22 points below the 97.95% reported on this dataset family [4]. Since

TABLE III

CROSS-LABEL DUPLICATE GROUPS AT TWO GRANULARITIES, OVER THE SAME BYTE-IDENTICAL GROUPS.

	Six categories	Three zones
Conflicted duplicate groups	119	103
Conflicted files	327	280
<i>Conflicted files by class</i>		
Positive / neutral	5	1
Negative affect	292	279
Surprise (unmapped)	30	—

1-NN upper-bounds what memorising a contaminated training set can buy, the published figures are not reachable by duplicate leakage under this feature representation. They must originate elsewhere—in augmentation applied before splitting in downstream derivatives, in test sets of a few dozen images, or in selection among many runs. Our audit constrains the explanation without completing it, and we flag this as the clearest target for follow-up work.

V. WHERE THE CONTRADICTIONS LIVE

The contradictions are not spread evenly. Mapping the vendor’s categories onto three valence/arousal zones—zone 1 (natural, joy), zone 2 (anger, fear), zone 3 (sadness), with surprise unmapped—and recounting gives Table III.

Zones 2 and 3 cover most of the archive, so some concentration is expected for free. We therefore test against a null that permutes labels across the zoneable images, preserving both the zone marginals and the duplicate-group structure, over 2,000 replicates. The null puts 59.5% of conflicted files in negative affect (95th percentile 61.7%). The observed share is **99.6%**, reached by no replicate ($p < 0.0005$).

The same null is informative in the other direction: it expects 535.8 conflicted files where the archive has 280. So the archive’s label assignments are *far more consistent than chance overall*, and almost all of the residual inconsistency lies inside negative valence. Whatever process filed these images separates positive from negative reliably and fails to separate the negative categories from one another.

We deliberately do not characterise that process as human annotator disagreement. The archive is a scrape with no documented labelling procedure, and a simpler explanation is available: the same image retrieved under two search queries and written into two folders. Either way the consequence for a model is identical—the label space as instantiated does not separate negative categories—and it is the consequence, not the mechanism, that our evidence supports. What the pattern rules out is that the inconsistency is generic noise.

VI. BEYOND FACE-ONLY CATEGORICAL RECOGNITION

Two design consequences follow, and one caution.

Coarser, action-oriented labels. If the reproducible distinction is valence and the irreproducible one is which negative category applies, a label space built on the reproducible boundary is better founded than one built across it. This is the argument for Emotion Zones: classes aligned with what a caregiver would do—de-escalate, engage, comfort—rather than with which Ekman category a rater would select. On this archive that is not a stylistic preference but a measured property: the zone boundary is where filing is consistent (Sec. V) and where protocol variance falls by 29.3% (Sec. IV). We have not run a caregiver study, so the *actionability* of the zones remains a design argument; only their statistical stability is demonstrated here.

The caution: never credit the chance floor. A collapse from six classes to three raises chance from 1/6 to 1/3, and comparing raw accuracies across the two silently rewards that. Our own earlier measurements show a commercial classifier moving from 28.17% (six categories) to 45.12% (three zones), an apparent 16.95-point gain; chance-corrected, the same relabeling is worth 3.88 points. We report chance-corrected figures for every cross-granularity comparison in this paper for exactly this reason.

Context beyond the face. Even at its cleanest, this task saturates well below deployment quality: the group-aware ceiling is 0.727 under zones, and Barrett *et al.* [22] argue facial configuration underdetermines emotional state in principle. The remaining headroom is unlikely to be in better face classifiers.

This is where multimodal and vision-language approaches earn their place, and the reason is a data argument rather than an accuracy argument. Training a face-only autism-specific model requires autism-specific face data, which is precisely the resource whose acquisition produced the retractions cited above. A vision-language model brings broad prior knowledge about scenes, activities, and posture without requiring any autism-specific training corpus, and can supply the environmental and interaction context a cropped face cannot. Reporting a zone together with an explanation a caregiver can check is also more auditable than a category with a confidence score—which matters given that any such system is, on this evidence, being evaluated on instruments that do not yet support the claims made for them.

VII. LIMITATIONS AND ETHICS

This audits one archive and does not establish prevalence, though the qualitative findings of [18] are consistent. Duplicate detection is exact-hash and perceptual-hash based and will miss re-encoded or cropped duplicates, so our rates are lower bounds; we did not validate the Hamming-4 threshold against hand-labelled pairs. The archive supplies no subject identities, so our groups are duplicate and augmentation families, not people; a subject-disjoint split would be stricter. Our features come from a web-pretrained backbone, so we cannot exclude that the encoder has previously seen these images—a caveat that applies to every published result on this archive and that

our “0% leakage” rows do not escape. Finally, the memorisation ceiling bounds contamination under *these* features; a fully fine-tuned network was not run.

Ethics. This is secondary analysis of a public dataset with no interaction with human subjects and no attempt to re-identify anyone; it is not human-subjects research under 45 CFR 46.102, and no IRB determination was required. We report only aggregate statistics: no image, filename, hash, or per-image prediction is released. The image corpus was deleted after hashing and feature extraction. We make no claim that any depicted child is autistic, that any label reflects a felt emotion, or that any system should be deployed on this basis; where Sec. V references findings about autistic expression, it does so to motivate a hypothesis, not to assert that this archive depicts diagnosed children. The archive’s provenance is disputed and its license unstated, so we redistribute nothing.

VIII. CONCLUSION

A widely used public autism emotion archive ships two overlapping copies sharing 45% of the smaller, is 21.4% duplicated, and contains 119 groups of pixel-identical images with contradictory labels. Varying only the split protocol moves five-category balanced accuracy across a 40.4-point band, so a single number on this archive carries little information about a model. Duplicate leakage is not the main culprit—worth 5.0 points, and only 2.1 to an explicit memoriser—and no protocol reaches the published figures, which we therefore cannot attribute to contamination.

The label contradictions concentrate in negative affect at 99.6% against 59.5% expected under a marginal-preserving null, while the archive is otherwise more self-consistent than chance. The categorical schema, not generic noise, is where the inconsistency lives, and the three-zone collapse cuts chance-corrected protocol variance by 29.3%. Coarser action-oriented zones and multimodal context are better founded responses than finer face-only categories. Any future benchmark here should report a duplicate audit, a group-aware split, and chance-corrected metrics as a condition of being believed—and, given how this archive was assembled, should not be built by scraping.

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REFERENCES

- [1] K. A. Shaw *et al.*, “Prevalence and early identification of autism spectrum disorder among children aged 4 and 8 years,” *MMWR Surveillance Summaries*, vol. 74, no. 2, Centers for Disease Control and Prevention, 2025.
- [2] National Autistic Society, “The double empathy problem,” <https://www.autism.org.uk/advice-and-guidance/professional-practice/double-empathy>, 2018.
- [3] P. Radočaj and G. Martinović, “Emotion recognition in autistic children through facial expressions using advanced deep learning architectures,” *Applied Sciences*, vol. 15, no. 17, p. 9555, 2025.

- [4] S. Hanumantharayappa and M. R. Bharamagoudra, "Confidence-weighted ensemble learning and graph-based shapley theory for refined emotion prediction in autistic children," *Journal of Trends in Computer Science and Smart Technology*, vol. 8, no. 3, pp. 745–767, 2026.
- [5] D. Kiejdo, M. Depka Prądzinska, and T. Zawadzka, "Late fusion model for emotion recognition from facial expressions and biosignals in a dataset of children with autism spectrum disorder," *Sensors*, vol. 25, no. 24, p. 7485, 2025.
- [6] H. Kalantarian, K. Jedoui, K. Dunlap, J. Schwartz, P. Washington, A. Husic, Q. Tariq, M. Ning, A. Kline, and D. P. Wall, "The performance of emotion classifiers for children with parent-reported autism: Quantitative feasibility study," *JMIR Mental Health*, vol. 7, no. 4, p. e13174, 2020.
- [7] C. Grossard, A. Dapogny, D. Cohen, S. Bernheim, E. Juillet, F. Hamel, S. Hun, J. Bourgeois, H. Pellerin, S. Serret, K. Bailly, and L. Chaby, "Children with autism spectrum disorder produce more ambiguous and less socially meaningful facial expressions: an experimental study using random forest classifiers," *Molecular Autism*, vol. 11, p. 5, 2020.
- [8] F. M. Talaat, "Retracted article: Real-time facial emotion recognition model based on kernel autoencoder and convolutional neural network for autism children," *Soft Computing*, 2023, retracted 16 November 2025.
- [9] İ. Güngör, A. Sarman, and S. Tuncay, "Retracted: Can artificial intelligence and face recognition using deep learning detect emotions in children with autism?" *PLOS ONE*, 2025, retracted 23 December 2025.
- [10] F. X. Gaya-Morey, S. Ramis, J. M. Buades-Rubio, and C. Manresa-Yee, "Assessing the efficacy of deep learning approaches for facial expression recognition in individuals with intellectual disabilities," *Multimedia Systems*, 2026.
- [11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.
- [12] B. Barz and J. Denzler, "Do we train on test data? purging CIFAR of near-duplicates," *Journal of Imaging*, vol. 6, no. 6, p. 41, 2020.
- [13] C. G. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.
- [14] B. Recht, R. Roelofs, L. Schmidt, and V. Shankar, "Do ImageNet classifiers generalize to ImageNet?" in *Proc. Int. Conf. Machine Learning (ICML)*, 2019.
- [15] P. Ramos, R. Ramos, and N. Garcia, "Data leakage in visual datasets," in *Proc. IEEE/CVF Int. Conf. Computer Vision Workshops (ICCVW)*, 2025, arXiv:2508.17416.
- [16] Y. K. Adimoolam, C. Poullis, and M. Averkiou, "Data leakage detection and de-duplication in large scale geospatial image datasets," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [17] A. Apicella, F. Isgrò, and R. Prevete, "Don't push the button! exploring data leakage risks in machine learning and transfer learning," *Artificial Intelligence Review*, 2025.
- [18] R. Chaiti *et al.*, "Emotion recognition of autistic children from facial images using hybrid model," in *Proc. IEEE Int. Conf.*, 2024.
- [19] A. Birhane and V. U. Prabhu, "Large image datasets: A pyrrhic win for computer vision?" in *Proc. IEEE Winter Conf. Applications of Computer Vision (WACV)*, 2021.
- [20] T. Gebu, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.
- [21] A. Paullada, I. D. Raji, E. M. Bender, E. Denton, and A. Hanna, "Data and its (dis)contents: A survey of dataset development and use in machine learning research," *Patterns*, vol. 2, no. 11, p. 100336, 2021.
- [22] L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, and S. D. Pollak, "Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements," *Psychological Science in the Public Interest*, vol. 20, no. 1, pp. 1–68, 2019.
- [23] C. T. Keating, S. Sowden-Carvalho, H. O'Donoghue, and J. L. Cook, "Mismatching expressions: Spatiotemporal and kinematic differences in autistic and non-autistic facial expressions," *Autism Research*, vol. 19, no. 2, p. e70157, 2026.
- [24] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [25] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk *et al.*, "Accounting for variance in machine learning benchmarks," in *Proc. Machine Learning and Systems (MLSys)*, 2021.